INCH-POUND

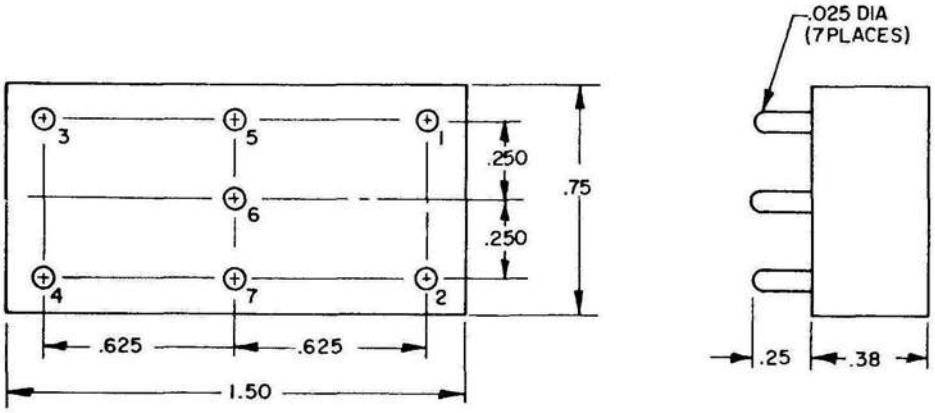
MIL-PRF-27/27A
28 April 2008
SUPERSEDING
MIL-PRF-27/27
15 September 1975

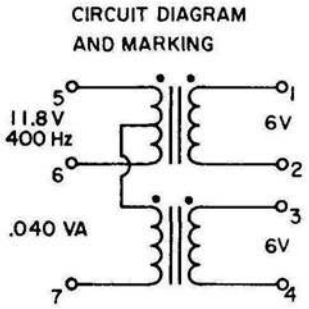
PERFORMANCE SPECIFICATION SHEET

TRANSFORMER, POWER, STEP DOWN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.





INCHES	MM
.025	.64
.250	6.35
.38	9.65
.625	15.88
.75	19.05
1.50	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm) and ±0.02 (0.51 mm) for remaining dimensions.
4. Marking shall be on the top of the case.
5. Terminals (5-6-7) may be to synchro input and terminals (1-2-3) may be to resolver output.

FIGURE 1. Dimensions and configurations

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage:
(5-6-7) – 11.8 volts, 400 Hz, 0.0035 ampere maximum.

Secondary voltage:
(1-2) – 6 volts, .6 mA rms.
(3-4) – 6 volts, .6 mA rms.

Working voltage:
(5-6-7) – 18 volts maximum.
(1-2) (3-4) – 18 volts maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle. Continuous.

Case: Encapsulated.

Material: Diallyl phthalate, glass filled.

Terminals: Pin type, copper clad, steel, tinned, copper weld.

Terminal height: .25 inch, minimum.

Terminal strength: Method 211, MIL-STD-202, test condition A, 2.0 pounds.

Dielectric withstanding voltage (at sea level): 50 volts rms.

Electrical characteristics:

No load: With 11.8 volts, 400 Hz across (5-6-7), current in (5-6-7) shall not exceed 3 mA and the power in (5-6-7) shall not exceed 20 milliwatts.
Voltage across (1-2) and (3-4) shall be 6 volts, respectively.

Rated load: With 11.8 volts, 400 Hz; .0035 ampere, maximum across (5-6-7).
Voltage across (1-2) and (3-4) shall be 6 volts, respectively.

Shaft angle transformation accuracy: 5 minutes of arc synchro input to resolver output.

Temperature rise: 5°C maximum.

Marking location: See figure 1.

Part or Identification Number (PIN): M27/27-01.

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air force – 11
DLA-CC

Preparing activity:
DLA - CC

(Project 5950-2008-005)

Review activities:
Army – AR, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at http://assist.daps.dla.mil.

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